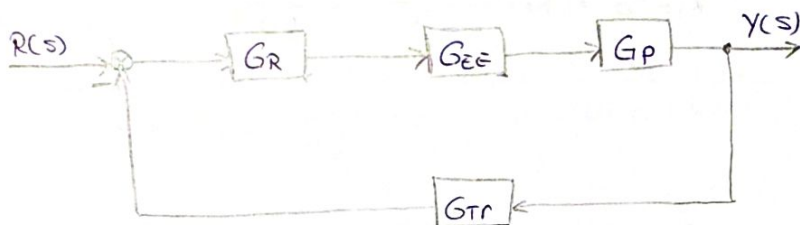


$$a) G_p = \frac{20}{s+1} = \frac{20}{1 + \frac{20K_2}{s+1}} = \frac{20}{s+1+20K_2}$$



$$G_R(s) = \frac{K_1}{s}$$

$$G_{EE}(s) = 1 = G_{TR}(s)$$

b) $K_1, K_2 = ?$ $K_V = 20$; $\zeta = 0,5$

$$G_e = \frac{G_R \cdot G_{EE} \cdot G_P}{1 + G_R \cdot G_{EE} \cdot G_P \cdot G_{TR}} = \frac{\frac{K_1}{s} \cdot \frac{20}{s+1+20K_2}}{1 + \frac{K_1}{s} \cdot \frac{20}{s+1+20K_2}} = \frac{\frac{20K_1}{s(s+1+20K_2)}}{\frac{s(s+1+20K_2) + 20K_1}{s(s+1+20K_2)}} =$$

$$= \frac{20K_1}{s^2 + s(1+20K_2) + 20K_1} = \frac{K \cdot \omega_m^2}{s^2 + 2\zeta\omega_m s + \omega_m^2} \Rightarrow \begin{aligned} K &= 1 \\ \omega_m^2 &= 20K_1 \\ 2\zeta\omega_m &= 1+20K_2 \\ \zeta &= \frac{1+20K_2}{2\sqrt{20K_1}} \Rightarrow \end{aligned}$$

$$\Rightarrow \sqrt{20K_1} = 1+20K_2$$

$$K_V = \lim_{s \rightarrow 0} s \cdot G_R \cdot G_{EE} \cdot G_P \cdot G_{TR} = \lim_{s \rightarrow 0} s \cdot \frac{20K_1}{s(s+1+20K_2)} = \frac{20K_1}{1+20K_2}$$

$$\frac{20K_1}{1+20K_2} = 20 \Rightarrow 20K_1 = 20 + 400K_2$$

$$\begin{cases} 1 + 20K_2 = \sqrt{20K_1} \\ 20 + 400K_2 = 20K_1 \Rightarrow K_2 = \frac{20K_1 - 20}{400} = \frac{K_1 - 1}{20} \end{cases}$$

$$1 + 20 \cdot \frac{K_1 - 1}{20} = \sqrt{20K_1}$$

$$1 + K_1 - 1 = \sqrt{20K_1}$$

$$K_1 = \sqrt{20K_1} \quad |^2$$

$$K_1^2 = 20K_1 \Rightarrow K_1^2 - 20K_1 = 0$$

$$K_1(K_1 - 20) = 0 \Rightarrow K_1 = 0$$

$$K_1 = 20 \Rightarrow K_2 = \frac{20 - 1}{20} = \frac{19}{20}$$

c) t_v , M_v , t_s

$$t_v = \frac{\ddot{u}}{\omega_m \sqrt{1 - \zeta^2}}$$

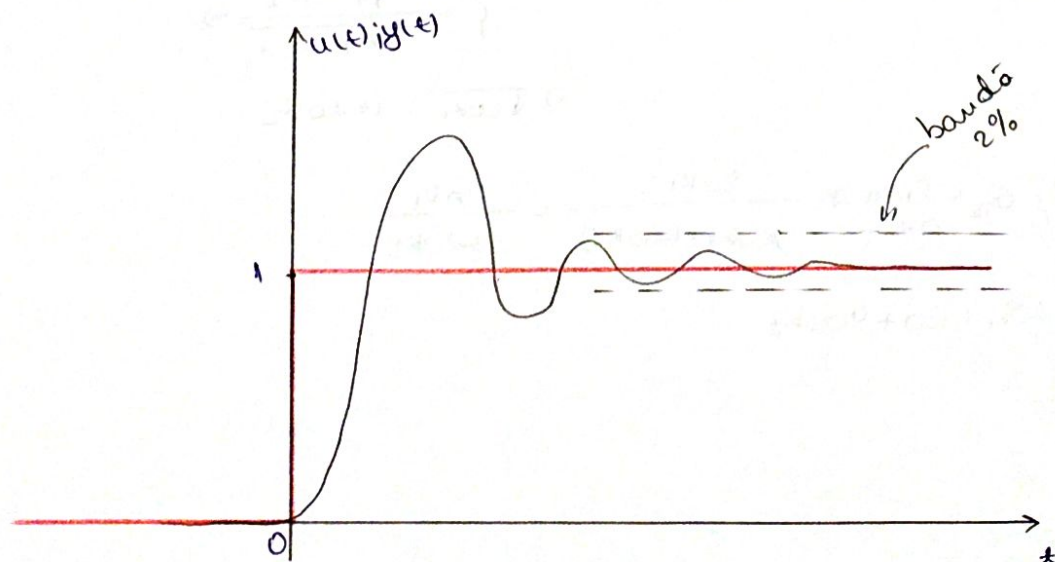
$$\omega_m = \sqrt{20K_1} = \sqrt{20 \cdot 20} = 20$$

$$t_v = \frac{\ddot{u}}{20 \sqrt{1 - (0,5)^2}} = \frac{\ddot{u}}{20 \sqrt{0,75}} = \frac{\ddot{u}}{20 \cdot 0,87} = \frac{\ddot{u}}{17,32} = 0,18$$

$$M_v = e^{-\frac{\ddot{u} \zeta}{\sqrt{1 - \zeta^2}}} = e^{-\frac{\ddot{u} \cdot 0,5}{\sqrt{1 - (0,5)^2}}} = e^{-\frac{1,5 \ddot{u}}{0,87}} = e^{-1,80} = 0,16$$

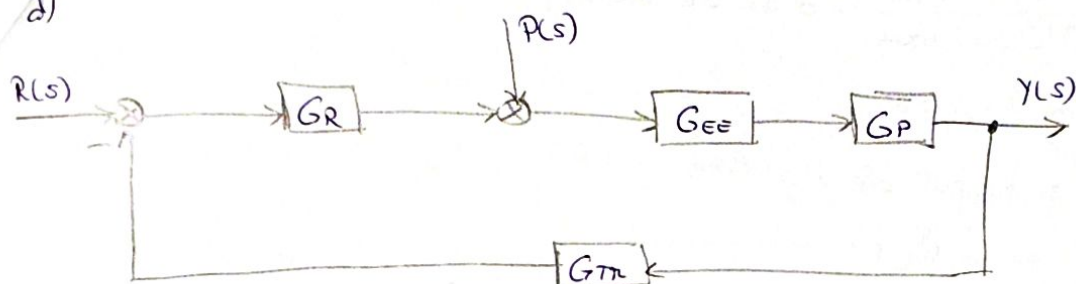
$$t_s = \frac{4}{\zeta \omega_m} = \frac{4}{20 \cdot 0,5} = \frac{4}{10} = 0,4 \text{ (2\%)}$$

$$t_s = \frac{5}{\zeta \omega_m} = \frac{5}{20 \cdot 0,5} = \frac{5}{10} = 0,5 \text{ (5\%)}$$



TS - +

d)



$$G_{op} = \frac{G_{EE} \cdot G_P \cdot G_{TR}}{1 + G_{EE} \cdot G_P \cdot G_{TR} \cdot G_R} = \frac{\frac{20}{s+1+20K_2}}{1 + \frac{20}{s+1+20K_2} \cdot \frac{K_1}{s}} = \frac{20A}{s(s+1+20K_2)+20K_1}$$

$$= \frac{20}{s(s+1+20 \cdot \frac{19}{20})+20 \cdot 20} = \frac{20}{s^2 + 20s + 400}$$

- e) - adăugarea unui pol suplimentar mai aproape de origine decât poli dominanta → micșorarea suprareglajului
poli dominanta $p_{1,2} = -10 \pm 14,32i$ (calc în Matlab)
pol adăugat: $p = -1$

$$G = \frac{20K_1}{[s^2 + s(1+20K_2) + 20K_1](s+1)} = \frac{20 \cdot 20}{[s^2 + s(1+20 \cdot \frac{19}{20}) + 20 \cdot 20](s+1)}$$

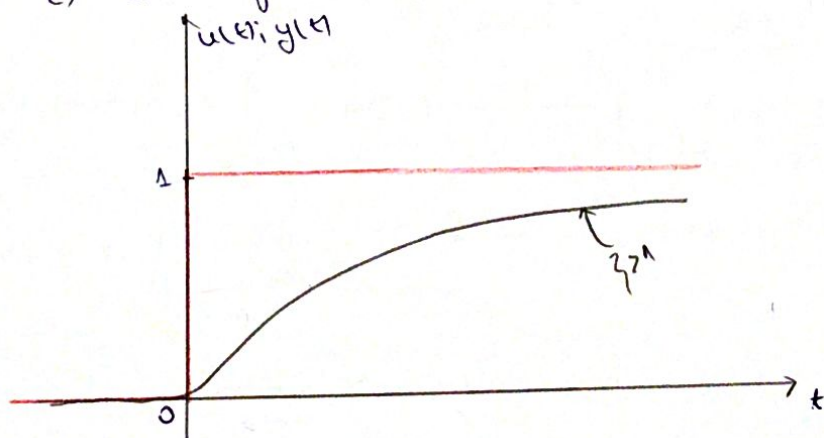
$$= \frac{400}{(s^2 + 20s + 400)(s+1)} = \frac{400}{s^3 + s^2 + 20s^2 + 20s + 400s + 400} = \frac{400}{s^3 + 21s^2 + 420s + 400}$$

② Teorie

a) $\frac{dI_+(t)}{dt} = f(t)$; $\int_0^t f(\tau) d\tau = I_+(t)$

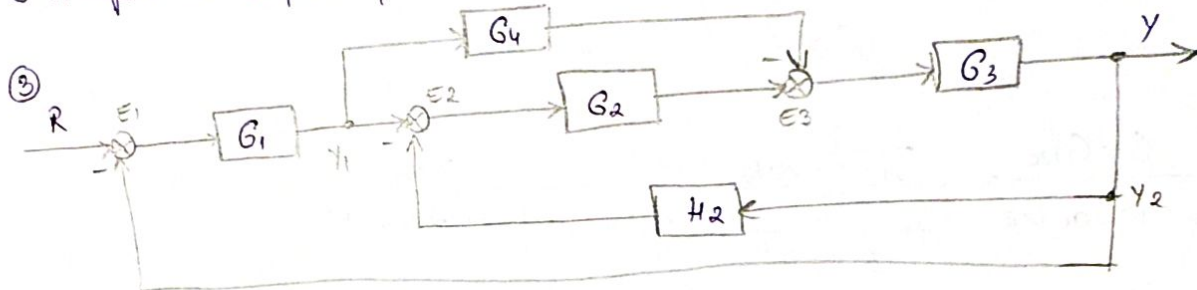
b) fc de transf a căui directe este $G(s)$

c) - sist supraamortizat: $\gamma > 1$

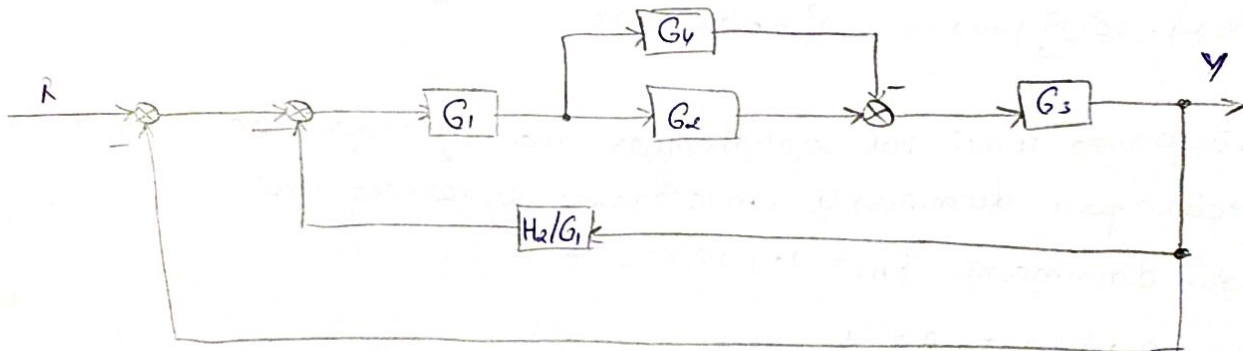


4. Fie $G(s) = \frac{b_0 s^m + b_1 s^{m-1} + \dots + b_m}{a_0 s^m + a_1 s^{m-1} + \dots + a_m}$ - o fc de transfer. Condiția de realizabilitate fizică este: $m \geq n$

5 Timpul de creștere și timpul de stabilire



• modificare de topologie:



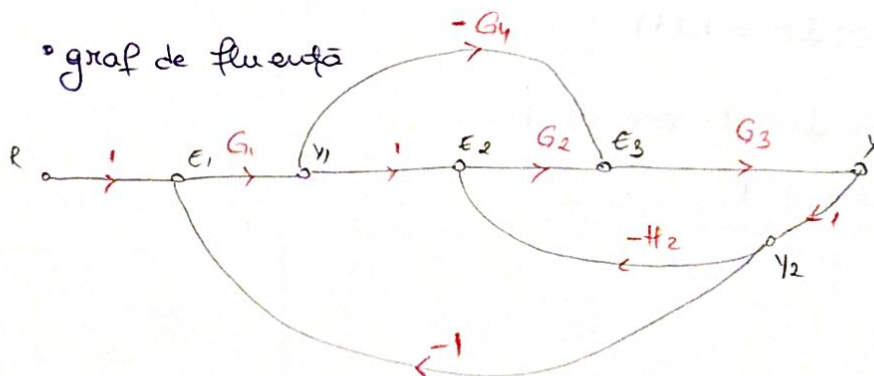
• G_2 și G_4 - paralel: $G_{e1} = G_2 - G_4$

• G_1 , G_{e1} și G_3 - serie: $G_{e2} = G_1 \cdot G_{e1} \cdot G_3$

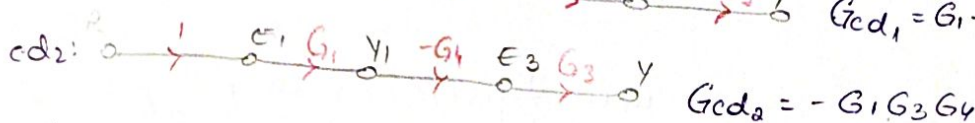
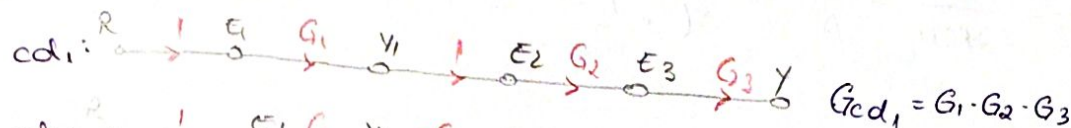
• H_2/G_1 și 1 - paralel: $G_{e3} = \frac{H_2}{G_1}$

• G_{e2} și G_{e3} - reacție: $G_e = \frac{G_{e2}}{1 + G_{e2} G_{e3}} = \frac{G_1 G_3 (G_2 - G_4)}{1 + \frac{G_1 G_3 (G_2 - G_4) \cdot H_2}{G_1}} = \frac{G_3 G_2 - G_3 G_4}{G_1 + G_1 G_3 G_2 H_2 - G_1 G_3 G_4 H_2}$

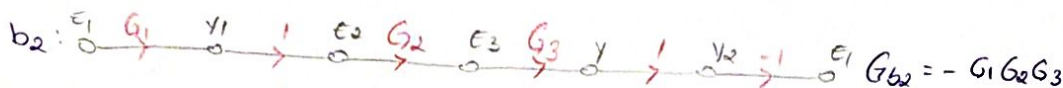
• graf de fluxuri



calea directă: 2.



bucle: 3



$$G_e(s) = \frac{1}{\Delta(s)} \cdot \sum_m G_{cdm}(s) \cdot \Delta_m(s)$$

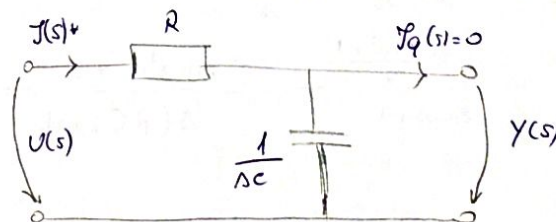
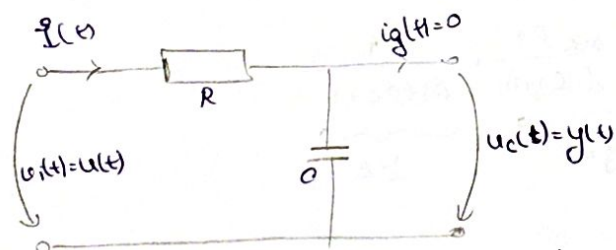
$$\Delta(s) = 1 - (G_{b1}(s) + G_{b2}(s) + G_{b3}(s))$$

$$\Delta_1(s) = 1 \quad ; \quad \Delta_2(s) = 1$$

$$G_e(s) = \frac{1}{1 - G_{b1}(s) + G_{b2}(s) + G_{b3}(s)} \cdot [G_{cd1} \cdot 1 + G_{cd2} \cdot 1] = \frac{G_1 G_2 G_3 - G_1 G_3 G_4}{1 + G_2 G_3 H_2 + G_1 G_2 G_3 - G_1 G_3 G_4}$$

- (4) - circuit RC cu ieșirea pe C
 - comd imit $y(0) = a$
 - rasp $u(t) = 1 + (e)$ în y_m și y_f

schema operațională



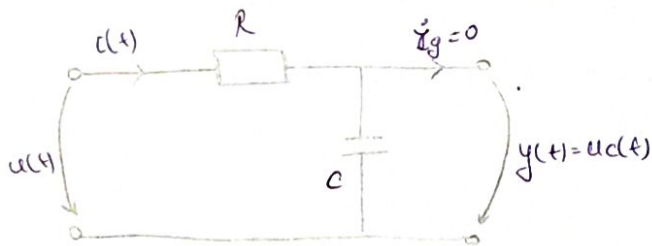
$$G(s) = \frac{Y(s)}{U(s)} = \frac{y_0 \cdot \frac{1}{sC}}{Y(s) \cdot R + Y(s) \cdot \frac{1}{sC}} = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = \frac{\frac{1}{sC}}{\frac{sRC + 1}{sC}} = \frac{1}{sRC + 1}$$

$$Y(s) = G(s) \cdot U(s) = \frac{1}{sRC + 1} \cdot \frac{1}{s} = \frac{C_1}{sRC + 1} + \frac{C_2}{s}$$

$$C_1 = (sRC + 1) Y(s) \Big|_{s = -\frac{1}{RC}} = \left(\frac{1}{sRC + 1} \cdot \frac{1}{s} \right) \Big|_{s = -\frac{1}{RC}} = \frac{1}{-\frac{1}{RC} + 1} = -RC$$

$$C_2 = s \cdot Y(s) \Big|_{s=0} = s \cdot \frac{1}{sRC + 1} \Big|_{s=0} = \frac{1}{sRC + 1} \Big|_{s=0} = 1$$

$$Y(s) = \frac{C_1}{sRC+1} + \frac{C_2}{s} = \frac{-RC}{sRC+1} + \frac{1}{s} = \frac{-\frac{RC}{RC}}{s + \frac{1}{RC}} + \frac{1}{s} \Rightarrow -e^{-\frac{t}{RC}} + 1 + (t) = y(t)$$



$$u_C(t) = \frac{1}{C} \int_0^t i(\tau) d\tau \Rightarrow i(t) = C \frac{du_C(t)}{dt}$$

$$u(t) = u_R(t) + y(t)$$

$$u_R(t) = i(t) \cdot R = RC \cdot \frac{du_C(t)}{dt} = RC \cdot \frac{dy(t)}{dt}$$

$$u(t) = RC \cdot \frac{dy(t)}{dt} + y(t)$$

$$\mathcal{L} \left\{ \frac{dy(t)}{dt} \right\} = sY(s) - y(0) = sY(s) - a$$

$$\mathcal{L} \{ y(t) \} = Y(s)$$

$$\mathcal{L} \{ u(t) \} = \frac{1}{s}$$

$$RC[sY(s) - a] + Y(s) = \frac{1}{s}$$

$$RCsY(s) - RCa + Y(s) = \frac{1}{s}$$

$$Y(s)[RCs + 1] = \frac{1}{s} + \frac{sRCa}{s}$$

$$Y(s) = \frac{\frac{sRCa + 1}{s}}{RCs + 1} = \frac{sRCa + 1}{s(RCs + 1)} = \underbrace{\frac{sRCa}{s(RCs + 1)}}_{y_M} + \underbrace{\frac{1}{s(RCs + 1)}}_{y_F}$$